



Machine edge-aware IoT framework for real-time health monitoring: Sensor fusion and AI-driven emergency response in decentralized networks

Asma Alshuhail^a, Amnah Alshahrani^b, Hany Mahgoub^c, Mukhtar Ghaleb^d,
Abdulbasit A. Darem^{e,*}, Nojood O. Aljehane^f, Modhawi Alotaibi^g, Fahad Alzahrani^h

^a Department of Information Systems, College of Computer Sciences & Information Technology, King Faisal University, Saudi Arabia

^b Department of Information Systems, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia

^c Department of Computer Science, Applied College at Mahayil, King Khalid University, Saudi Arabia

^d College of Computing and Information Technology, University of Bisha, Bisha 61922, Saudi Arabia

^e Center for Scientific Research and Entrepreneurship, Northern Border University, Arar 73213, Saudi Arabia

^f Department of Computer Science, Faculty of Computers and Information Technology, University of Tabuk, Tabuk, Saudi Arabia

^g Department of Computer Science, College of Computer Science and Engineering, Taibah University, Taibah, Saudi Arabia

^h Department of Information and Computer Science, College of Computing and Mathematics, King Fahad University of Petroleum and Minerals, Saudi Arabia

ARTICLE INFO

Keywords:

Health monitoring systems
Cloud infrastructure
IoT framework
Real-time sensor fusion
Edge-based decision system
Anomaly detectors
Wearable biosensors

ABSTRACT

Health monitoring systems require wider deployment to support medical institutions in their care of both chronic patients and elderly people while providing urgent emergency services. Systems based on traditional cloud infrastructure through central locations create multiple problems that include delays in performance along with failures in connectivity and restrictions in system usage, and privacy risks. The IoT framework resolves issues through real-time sensor fusion, where edge-based decision systems use lightweight AI anomaly detectors for urgent emergency choices. Several wearable biosensors analyze heart rate and blood oxygen saturation rates and body temperature information as they process fall metrics live. The edge nodes perform speedy AI-based analytics directly for essential health situations through their efficient processing capabilities to establish brief cloud system connections. In distributed networks, system-wide alerts are activated through the emergency alert protocol after distributed networks execute autonomous decision-making processes. The proposed design utilizes an adaptive mesh networking approach to ensure dependable transmission across diverse settings and support ongoing remote monitoring. A hybrid sensor fusion algorithm analyses different physiological parameters, such as ECG, SpO₂, and body temperature, to detect potentially dangerous signals and set off local emergency alarms. The figures show that the system achieves high detection accuracy (up to 95.4 %) within just 0.045 s and uses less power. The efficacy of the fall and cardiac event detection capabilities was shown in a simulation of real-life assisted living settings. The findings demonstrate that the system provides accurate, secure health data sharing in a scalable and rapid manner for dispersed IoT environments.

1. Introduction

Advanced healthcare solutions emerged through a combination of adaptive methodologies between Internet of Things (IoT) systems and edge computing, and artificial intelligence (AI) and biomedical sensing [1]. Modern medical practices now function as proactive health management because continuous vital health parameter tracking through interconnected smart devices exists. Real-time health monitoring systems help identify early signs of abnormal body alterations while

improving fast incident responses and easing stress on main healthcare institutions.

The implementation of big sensor network systems in critical healthcare settings faces ongoing challenges because of delayed communication and network traffic and security issues, and constrained computational capacity. An edge-aware Internet of Things system with real-time health surveillance features serves emergency responses by combining sensor output and processing decentralization techniques. The framework achieves speed reduction and user confidentiality

* Corresponding author.

E-mail address: basit.darem@nbu.edu.sa (A.A. Darem).

<https://doi.org/10.1016/j.aej.2025.08.030>

Received 8 May 2025; Received in revised form 12 August 2025; Accepted 19 August 2025

Available online 6 September 2025

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maintenance through its approach of assigning computational tasks to the nearest edge nodes. Sensor fusion procedures unite several types of physiological data, which includes electrocardiograms (ECG) with temperature readings along with blood oxygen saturation (SpO2) combined with motion measurements to meet stringent reliability and robustness requirements [2].

The framework's primary advantage lies in having an AI-powered anomaly detection algorithm operating directly inside edge devices. The autonomous engine employs neural networks together with statistical models for critical health event identification, then makes independent alerts and emergency service requests, which do not need cloud platform engagement.

The automated response system operates faster than anticipated to address emergency conditions, which include heart attacks along with hypoxia, and elderly persons [3]. Decentralized network architecture helps prevent system breakdown because it removes the failing system from conventional centralized systems.

The system uses data integrity methods based on blockchain principles to optimize distributed logging operations. The proposed scheme utilizes three nested system tiers, starting with the sensing layer that followed by the edge processing layer, and ending with the communication layer for secure MQTT and CoAP data transmission. The cohesive layer integration between elements maintains an ongoing secure health monitoring system, which provides scalability and operates with minimal power needs.

The main contributions of this study are summarized as follows:

1. Development of an edge-aware IoT framework integrating multi-sensor fusion, AI-driven anomaly detection, and decentralized emergency response.
2. Implementation of lightweight AI models optimized for embedded edge devices to achieve real-time decision-making without cloud dependence.
3. Design of a decentralized mesh networking protocol for resilient communication and adaptive path selection in dynamic health monitoring environments.
4. Demonstration of high accuracy, low latency, and energy efficiency in real-world simulated healthcare scenarios.

The remainder of this paper is organized as follows: [Section 2](#) reviews Literature Review and current advancements in IoT-based healthcare. [Section 3](#) details the proposed methodology. [Section 4](#) presents the experimental setup. [Section 5](#) discusses the results and comparative performance analysis. [Section 6](#) concludes the study and outlines future research directions.

2. Literature review

Internet of Things (IoT), together with artificial intelligence (AI) and edge computing systems, has created modern smart healthcare systems that provide real-time personalized, proactive patient monitoring solutions. Health monitoring platforms need to become reliable and operate with low latency because chronic illnesses, along with an aging population, make such services important. Intelligent wearable devices track ongoing physiological patterns effectively while facing several practical obstacles involving main processing, installation barriers, and emergency time delays.

The following section evaluates the development and connectivity of advanced technologies that serve as basic infrastructure for upcoming healthcare solutions. The digital health sector experienced transformation through wearable technology, which provides ongoing non-traumatic assessment of essential medical parameters [4]. Modern wearable technology includes a diverse set of biosensors, including electrocardiograms (ECGs) for heart monitoring and pulse oximeters (SpO₂) for oxygen monitoring, as well as thermistors and inertial measurement units (IMUs), which track heart rate and physical movements

and risks of falling [5].

Mobile applications wirelessly receive sensor data, which enables users to view it and physicians and caregivers to support decision-making processes. Two crucial boundaries emerge from cloud-based platforms in these systems: they cause transmission delays in data transfer and create vulnerabilities for protected patient data, which threatens privacy. The medical IoT system requires sensor fusion techniques to integrate multiple data sources from physiological sensors to achieve better results and avoid failure, and to enhance contextual understanding [6,30].

The analysis of combined ECG measurements with body motion data results in better detection abilities of arrhythmias and syncopal events. The majority of current systems perform sensor fusion tasks through operations conducted either on cloud platforms or centralized computing servers. The proposed architecture acts as a limit for vital healthcare applications because it creates delays while requiring secure internet connections [7].

AI-based health anomaly detection models trained to identify cardiac arrhythmias and detect apnea or the early indicators of stroke function from cloud platforms [8]. The design is ineffective for emergencies requiring real-time decisions because it lacks the speed to respond within milliseconds. Edge computing provides strong operational potential through its ability to process health data on edge devices such as microcontrollers, embedded systems, and wearable gateways near the source. Real-time responsiveness becomes achievable through this technological paradigm because it eliminates the need for continuous cloud access, which remains vital for life-threatening situations such as cardiac arrest or stroke onset [9].

Recent investigations have introduced lightweight AI models consisting of decision trees together with shallow neural networks and compressed convolutional models capable of efficient deployment on devices with limited resources [10,26,29]. The models active researchers are studying to determine their generalizability for dynamic health situations while improving their overall robustness between ambulatory patients and vitals with fluctuating patterns.

Health professionals address the constraints of centralized health architectures through decentralized network development, which employs peer-to-peer communication protocols [11]. The infrastructure supports automatic redundancy as well as fault tolerance and scalability, which makes it suitable for large-scale distributed health monitoring systems. However, effective deployment in medical contexts requires secure synchronization mechanisms, blockchain-based trust models, or reputation-based systems to ensure data integrity, confidentiality, and verifiability [12,28].

Emerging hybrid systems that combine edge intelligence, sensor fusion, and distributed emergency management are being verified in specific domains. Applications include fall detection for elderly populations [13], early detection of stroke indicators through multimodal sensing, and ambulance-based real-time telemetry systems that assist first responders. Yet, these systems often remain domain-specific and fragmented, lacking a unified architecture that can support generalized, multi-condition health monitoring with dynamic AI-driven response protocols [14,27].

To bridge these gaps, a comprehensive and unified edge-aware IoT framework is proposed. This system integrates multi-sensor fusion, localized AI inference, and decentralized emergency activation to enable scalable, responsive, and secure health monitoring across diverse environments, whether in homes, ambulances, or remote clinical setups [15].

[Table 1](#) summarizes some of the emerging technologies in the field of IoT-integrated applications.

2.1. Related work and comparative analysis

There are multiple works that are trying to merge IoT, edge computing, and AI to use in healthcare. To give an example, Hosseini

Table 1
Technological advancements in IoT-based applications.

Technology	Features	Limitations	Fusion Type	Processing Model
Smart Wristbands	Real-time ECG, HR monitoring	Poor battery life	Single-modal	Centralized
Smart Helmets	Impact sensors, GPS	Limited physiological monitoring	Multi-modal	Edge
Smart Beds	Sleep pattern, weight monitoring	High cost	Multi-modal	Centralized
Fall Detection Systems	Accelerometer-based	High false positives	Uni-modal	Local
AI-powered Health Apps	Predictive analysis	Delayed cloud response	Multi-modal	Cloud-based
Mobile ECG Devices	Real-time ECG visualization	Manual analysis required	Single-modal	Mobile edge
IoT-enabled Ambulances	GPS tracking, real-time vitals	Expensive deployment	Multi-modal	Hybrid
Distributed Sensor Networks	Redundant sensing, resilience	Complex synchronization	Multi-modal	Decentralized
Smart Inhalers	Dosage tracking, reminders	Non-real-time feedback	Single-modal	Edge
Post-surgical Monitors	Real-time wound temperature	Short battery cycle	Multi-modal	Hybrid

et al. [2] have offered a fall detection system that was implemented using one IMU, but it did not involve multi-modal integration of the physiological data. Moghadas et al. [9] devised a cardiac monitoring system based on fog computing, but that relied on a fixed internet connection, so it could not be used in low-resource settings. The research aimed at the effective use of deep learning inference on edge devices by Shuvo et al. [13], however, did not involve any decentralized emergency response mechanisms. Conversely, the proposed framework can be embedded with multi-sensor fusion, edge-local AI inference, and a decentralized mesh communication protocol, which allows them to be very responsive and provide high-level fault tolerance. Our approach can be juxtaposed with these previous works to argue that it maintains a consistent architecture with the ability to manage various health conditions in real-time, even when the network is unfavourable.

3. Methodology

The proposed framework integrates edge-level data acquisition, local AI processing, and decentralized communication to facilitate real-time health monitoring and emergency response.

The visual representation of the three-layer system layout is displayed in the graphical abstract (Fig. 1). The patient-worn devices at the bottom layer house different physiological sensors that gather information about heart rate, together with SpO₂ and body temperature. An edge device connects these devices through wireless transmissions. Between the edge node and the top layer exist two simultaneous data streams that direct information toward either an artificial intelligence evaluation engine or secure communication transmission. Data synchronization plus emergency trigger management occurs on the last layer through decentralized nodes that interconnect using a mesh topology. Visual markers display operational features, including swift response capabilities and sensor information merging, and network distribution capabilities. The research presents an extensive exploration of present-day studies, together with design approaches, hardware design, mathematical frameworks, and experimental proof, along with AI-based decision systems. Edge computing, along with decentralized

intelligence, enables the development of scalable privacy-aware smart healthcare systems within the expanding domain of health monitoring systems.

The methodology is structured around a modular hardware-software co-design that supports multi-sensor interfacing, on-device inference, and secure peer-to-peer data exchange. Physiological parameters such as heart rate, oxygen saturation, body temperature, and fall indicators are captured using a suite of wearable sensors. These raw signals undergo pre-processing and normalization before being passed through a sensor fusion engine that enhances reliability by mitigating sensor noise and data loss. A lightweight anomaly detection algorithm, optimized for execution on resource-constrained edge devices, evaluates deviations from baseline health metrics and triggers emergency responses when necessary. The distributed mesh network actively logs decisions that are selectively transferred for preserving data integrity and maintaining system stability.

$$HR = \frac{60}{RR} \quad (1)$$

Eq. (1) converts the time between two R-peaks of an ECG [16] waveform (RR interval) into heart rate in beats per minute (BPM). The RR interval reflects the duration between cardiac cycles and is a critical marker for arrhythmias, tachycardia, or bradycardia. In edge computing systems, where processing power is limited, this approach allows for straightforward implementation without the need for complex transformations. By utilizing basic peak detection algorithms to determine RR intervals, this formula supports continuous, low-latency monitoring of heart rate trends, which is particularly crucial during emergency conditions such as myocardial infarctions or cardiac arrests.

$$SpO_2 = 110 - 25 \cdot \left(\frac{AC_{red}/DC_{red}}{AC_{IR}/DC_{IR}} \right) \quad (2)$$

Eq. (2) [17] serves as a fundamental step for peripheral capillary oxygen saturation measurement through photoplethysmographic (PPG) signals acquired by wearable optical sensors. Non-pulsatile absorption is measured through DC, whereas the AC represents pulsatile absorption. The absorption levels of red and infrared light function differently according to the levels of tissue oxygenation. Users can obtain a dependable SpO₂ reading from the calculated ratio because assessment of peripheral capillary oxygen saturation is crucial for diagnosing COPD and COVID-19, among other respiratory conditions. Device-based logical processing performs hypoxia detection fast enough to warn patients before needing a server connection, thereby protecting patient safety in resource-limited settings.

$$T(^{\circ}C) = \frac{V_{sensor} - V_{offset}}{Slope} \quad (3)$$

Analog temperature sensors use voltage output as they change their voltage levels according to temperature variations, as in Eq. (3). The raw voltage gets transformed to degrees Celsius by applying calibration constants obtained through testing and the sensor documentation [18]. Monitoring body temperatures with accuracy helps physicians recognize

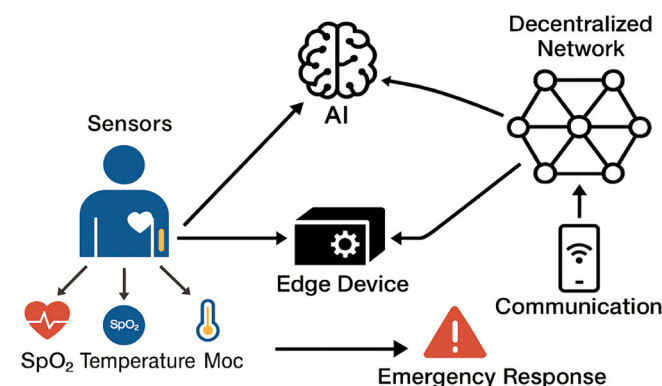


Fig. 1. Graphical representation of a health monitoring system.

early signs of fevers and hypothermia to control infections after surgery. The simple computational needs of this equation make it suitable for controller systems based on microcontrollers. Temperature functions as an important parameter that supports edge-based systems in performing complete health status inference alongside other vital measurements.

$$Fused_Output = \sum_{i=1}^n w_i \cdot S_i \quad (4)$$

Multiple sensor inputs, such as accelerometers and gyroscopes, as well as ECGs, merge their collective data into an aggregated output as in Eq. (4). The weighted sum method applies weights w_i to inputs S_i based on their reliability as well as noise and importance levels [19]. Health monitoring systems use sensor fusion to combine motion data with biometric signals, improving fall detection reliability and making physical activity detection more precise. Fusion technology both lowers unneeded alerts and improves understanding of contextual situations in elderly care systems and ambulatory monitoring environments that frequently experience data variations.

$$Score = \frac{\|x - \mu\|^2}{\sigma^2} \quad (5)$$

The Mahalanobis distance-derived statistical method in Eq. (5) calculates the degree of deviation for a new data point x from the mean μ through normalized variance σ^2 . The assessment functions to detect anomalies in real time for both ECG signals and pulse rates as well as body temperature measurements [20]. The detection system issues warnings when the obtained score reaches certain thresholds. This equation enables efficient computation and minimal memory use thus making it suitable for low-power edge systems as well as maintaining high sensitivity to critical health parameter deviations.

$$E = P \cdot t \quad (6)$$

Health devices that operate using batteries need to excel in energy efficiency characteristics. Eq. (6) finds the total energy usage by multiplying active time (t) with power consumption (P) [21]. The model enables practitioners to evaluate wearable battery capacity while aiding in power conservation by allowing adjustments such as power scaling, voltage, and scheduling device standby states. The energy consumption model helps developers maintain continuous device operation through challenging patient monitoring situations that require medical emergency transportation during times when charging facilities are unavailable.

$$L_{total} = L_{sensor} + L_{process} + L_{transmit} \quad (7)$$

The timely nature of real-time system delays becomes essential for

conversion along with efficient inference models perceiving these short-range wireless protocols reduce total system delay L_{total} in Eq. (7). The applications that require detection of falls or strokes need implementation delays of less than one second.

$$T_{emergency} = \theta_{HR} + \theta_{SpO2} + \theta_{Fall} \quad (8)$$

The rule-based emergency threshold in Eq. (8) integrates multiple vital signs into a composite score parameter. The weights θ for heart rate, oxygen saturation, and fall detection reflect their contribution to a medical crisis [23]. If the combined score crosses a predefined boundary, the system triggers emergency protocols such as alerting caregivers or activating GPS-based location broadcasting. This fusion-based threshold minimizes false positives and improves sensitivity. It is especially effective in environments where rapid triage decisions must be made without cloud support.

$$CR = \frac{Original\ Size}{Compressed\ Size} \quad (9)$$

Data compression is vital for reducing storage and transmission loads in health monitoring systems. Eq. (9) assesses how much a dataset can be reduced. In edge deployments, where bandwidth and storage are constrained, compression algorithms must be lightweight and real-time capable. A higher compression ratio translates to lower communication energy and faster transmission, which is critical in transmitting ECG or multi-sensor streams during emergencies [24]. Balancing compression with data integrity ensures that critical medical features are preserved post-compression.

$$Trust_i = \frac{Successful_Tx_i}{Total_Tx_i} \quad (10)$$

Trust mechanisms are crucial in decentralized, peer-to-peer health monitoring networks. This ratio-based metric in Eq. (10) evaluates the reliability of each node based on its history of successful transmissions [25]. A low trust score could indicate packet loss, malicious behavior, or hardware faults. The system can then reroute data through higher-trust nodes to maintain robustness. This mechanism enhances the scalability and fault-tolerance of decentralized systems, particularly in mission-critical deployments such as battlefield medicine, remote clinics, or home-based care for vulnerable patients.

The workflow of the Emergency Detection algorithm is adopted and implemented as mentioned in the following logic.

Algorithm. Emergency Detection(Input: SensorData)

```

Begin
  Normalize SensorData
  Compute FusedFeatures from SensorData
  Score ← CalculateAnomalyScore(FusedFeatures)
  If Score > Threshold then
    Trigger Emergency
  Else
    Continue Monitoring
End

```

emergency healthcare operations [22]. The complete system delay comprises acquisition delay through sensors (L_{sensor}), computational delay at the edge ($L_{process}$) as well as communication delay for data or alert transmission ($L_{transmit}$) as mentioned in Eq. (7). The optimization of each system component produces faster time frames needed for decision-making processes. All together faster analog-to-digital

Fig. 2 summarizes the methodology section, displaying a systematic way to approach the mentioned problem statement.

The sensor fusion algorithm combines ECG, SpO₂, and accelerometer data using a decision-level fusion approach that assigns appropriate weights to each source of information. Data from each sensor is cleaned and standardized before it is processed. Individual anomaly scores are

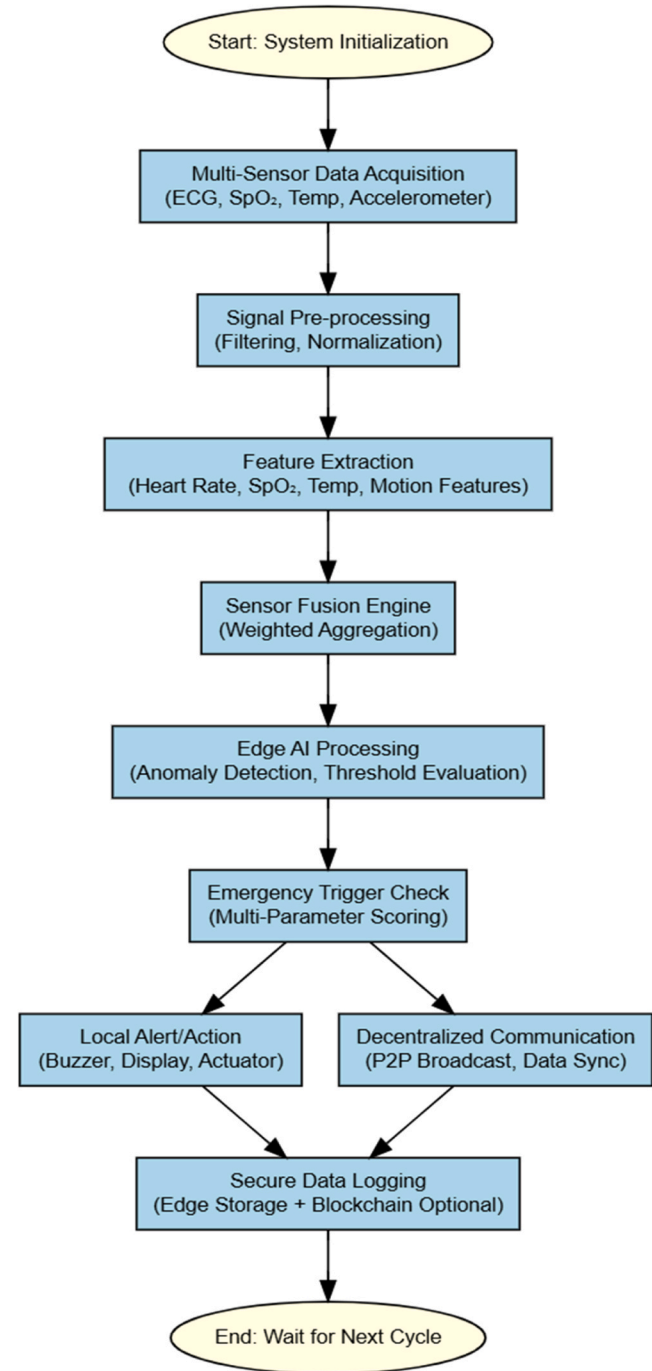


Fig. 2. Architecture diagram for the proposed Methodology.

calculated for each sensor. Health anomalies are detected by examining ECG parameters such as RR interval fluctuation and comparing SpO₂ data with oxygen saturation thresholds, while motion data is analyzed to identify unusual falls. Rules are used to determine how important each detected anomaly is with respect to its severity and reliability.

A simultaneous decrease in SpO₂ concentration and ECG anomalies will generate an immediate emergency call. This fusion method enhances the reliability of anomaly detection and enables the system to respond appropriately to situations in a timely and efficient manner. Further, the proposed framework utilizes a decentralized mesh networking approach to maintain uninterrupted communication among wearable nodes despite adverse environmental and connectivity changes. Mesh topology enables nodes to simultaneously broadcast and

receive information, ensuring improved tolerance to failures and scalability than fixed structures. When challenges such as signal interference, node disruptions, or limited transmission range occur, the system automatically finds and utilizes optimized paths by considering the condition of network links and available nodes. This approach maintains reliable communication as the network changes due to node movement or environmental variability in practical health tracking scenarios. The protocol showed resilience in situations where up to 25 % of nodes were unavailable. This framework is well-suited for applications in high-density networks found in homes, rural medical centres, and mobile emergency services due to the importance of maintaining continuous health monitoring and effective decision-making.

4. Experimental setup

To validate the proposed edge-aware IoT framework, a comprehensive experimental testbed was established, replicating real-world health monitoring conditions. The setup integrates multiple biomedical and motion sensors connected to an embedded edge device capable of real-time data acquisition, local processing, and decentralized communication. The hardware system was assembled to capture ECG signals, blood oxygen levels (SpO₂), skin temperature, and motion events such as falls using a triaxial accelerometer. Signal conditioning circuits, voltage regulation modules, and isolation components were employed to ensure clean signal acquisition and stable sensor operation. The central processing unit implements a low-power microcontroller with an edge AI board through which feature extraction joins anomaly detection, as well as decision-making algorithms operate directly on the device. The peer-to-peer wireless modules were configured to create decentralized mesh communication networks between multiple devices. A specialized container served as a form of wearable equipment for testing that subjects could wear on their body. System firmware achieved optimal energy efficiency through its timed execution of sensing, processing, and transmission responsibilities. The data captured by the system was recorded at the local node storage while simultaneously broadcasting to adjacent wireless network nodes to test data integrity. The experimental prototype requires several critical components with specified features that are detailed along with their implementation parameters in the following draft. Table 2 comprises a list of necessary components required to complete the setup

An experimental system consisted of a defined set of components for real-time edge-level physiological monitoring with low-power requirements. The fundamental component comprised a 32-bit ARM Cortex M4 microcontroller, which executed edge-level processing as it communicated with sensors using I²C and UART protocols because of its strong interrupt handling and average computational speed. An ECG sensor recording ECG data at a 1000 Hz rate monitored heart activity through an analog channel that ensured real-time heartbeat recognition for measuring heart rates. To measure arterial oxygen saturation, a dual-wavelength SpO₂ sensor was integrated using the I²C interface, enabling simultaneous transmission of red and infrared photoplethysmography

Table 2
Components necessary for experimental setup.

Item	Specification	Purpose	Interface
Microcontroller	32-bit ARM Cortex M4	Edge processing	I2C, UART
ECG Sensor	3-lead, 1000 Hz sampling	Heart rate	Analog
SpO2 Sensor	Dual wavelength	Oxygen monitoring	I2C
Accelerometer	3-axis, ±16 g	Fall detection	SPI
Temperature Sensor	0.1°C precision	Body temperature	Analog
Wireless Module	2.4 GHz, BLE 5.0	Communication	UART
Edge AI Engine	TinyML model	Anomaly detection	Internal
Battery	3.7 V Li-ion	Power supply	N/A

signals critical for oxygen level calculation. For motion detection and fall event analysis, a 3-axis accelerometer with a sensitivity range of ± 16 g was used, interfaced over SPI for fast and synchronized data transfer. Body temperature was recorded using an analog temperature sensor with 0.1°C precision, allowing accurate core temperature monitoring vital in clinical diagnostics.

Communication between devices was facilitated through a 2.4 GHz BLE 5.0 wireless module, connected over UART, to support mesh networking in a decentralized configuration with low power overhead. The core AI-driven decision logic was implemented using a TinyML model embedded within the microcontroller, forming the edge AI engine that performed anomaly detection and health condition scoring locally. A 3.7 V Li-ion battery served as the powering mechanism because it satisfied requirements for wearable technology by delivering efficient rechargeable stability. Multiple design components are integrated into this system to create a small format and reduce power usage while allowing independent healthcare observation without outside support of a 3.7 V Li-ion battery, which serves as the powering mechanism, as it satisfies requirements for wearable technology by delivering efficient rechargeable stability. Multiple design components are integrated into this system to create a small format and reduce power usage while allowing independent healthcare observation without outside support.

The proposed solution includes a portable ECG analysis engine that can identify major cardiac anomalies, including arrhythmias as they occur. The ECG Analytics module makes use of advanced algorithms to retrieve essential timeseries metrics such as the RR interval, HRV, and waveform pattern for further analysis. They are analyzed at the edge of the embedded microcontroller to detect conditions like bradycardia, tachycardia, and PVCs within the ECG signal. A customized anomaly detection algorithm properly adjusted using ECG analytics data flags unusual rhythms and signals an alert, obviating the need for cloud processing. This allows for swift and timely identification of potentially dangerous cardiac events.

The architecture also provides built-in flexibility for incorporating lightweight machine learning models to further enhance cardiac diagnosis in conditions that present a wide range of physiological and noise characteristics [31]. With this edge-computing solution, individuals in rural areas, the elderly, and people with a higher probability of having heart disease can benefit from real-time monitoring and prompt medical attention.

The framework is built with scalability in mind, making it able to handle small-scale remote patient monitoring or large-scale monitoring in hospitals, nursing homes, or rural areas. Edge models are maintained using periodic over-the-air (OTA) updates via encrypted channels. For personalized monitoring, incremental learning is supported through local dataset accumulation and retraining at designated intervals. These features ensure adaptability to individual user patterns and health baselines. Strong security and privacy protections are implemented by using AES-256 encryption during all communications between devices, implementing secure key exchange methods, and anonymizing sensitive patient data at the medical device's edge.

To ensure compliance with healthcare data standards such as HIPAA, the system only transports raw identifiable data to cloud services if the user has given explicit permission. The system's centralized network avoids relying on a central server for communication, distributing the information exchange across nodes, and improving network responsiveness. Individual edge devices can make independent judgments and instantly relay urgent messages to other sensors in the network. The flexible, modular nature of the sensor nodes and smart resource allocation approaches allows the system to handle increasing numbers of devices or coverage areas gracefully.

Layers of hardware and software architecture were adopted to realize the IoT system. The ARM Cortex M4 microcontroller at the sensing layer was interfaced with biomedical and motion sensors. The edge AI engine, based on a TinyML-based anomaly detection model, operated in real time on the extracted features of the ECG, SpO_2 ,

accelerometer, and temperature data. The AI-based decision logic was fine-tuned such that the processing latency was < 100 ms so that emergent events could be detected and communicated in real time. A BLE 5.0 mesh network with a peer-to-peer data communication protocol was also implemented at the communication layer to provide redundancy and fault tolerance. OTA updates guaranteed that the models of edge AI were timely without having to stop the systems.

5. Results and discussion

A series of tests on the proposed framework evaluated its performance in real-time health surveillance as well as network-based anomaly detection and fast emergency response functions for decentralized networks. Sensor data from the integrated sensor suite encompassing ECG readings, together with SpO_2 measurements and body temperature and movement activities, was obtained in diverse normal and abnormal health states. The edge AI system underwent testing to determine its capability for processing live physiological data until it detected aberrant patterns that would activate warning alerts swiftly. Multiple types of graphs showed the outputs from sensors and trends of features and system performance indicators like response times and energy consumption. Testing evaluated the reliability, together with the accuracy and efficiency of calculations performed at the edge alongside the decision mechanism operations. The proposed architecture will be evaluated through both simulations alongside graphical analyses, where matching tables demonstrate the combination of strengths and weaknesses across different evaluation measures.

The electrocardiogram (ECG) signal in Fig. 3 achieves capturing at 1000 Hz, which delivers fine resolution for identifying P waves for atrial depolarization and QRS complexes for ventricular depolarization alongside T waves for ventricular repolarization. The assessment of cardinal heart features relies heavily on these essential components. Consecutive R-wave peaks generate the R-R interval that leads to the instantaneous computation of heart rate (HR). The system performs ECG analysis at the edge node, which enables immediate arrhythmia detection through operations free from cloud dependence. The waveform clarity enables the use of segmentation algorithms together with artificial intelligence models to perform predictive diagnostics tests in both home and mobile patient monitoring systems.

The illustration presented in Fig. 4 depicts unfiltered and filtered photoplethysmography (PPG) signals for peripheral blood oxygen saturation (SpO_2) measurement. The peripheral blood oxygen saturation (SpO_2) determination depends on the signals detected from the photoplethysmography signals. Low-frequency variations along with high-frequency noise appear in raw waveforms because of user movements and suboptimal sensor placement, and sunlight interference. The Gaussian smoothing filter enhances peak consistency while improving periodicity in the signals. The system requires an improvement for accurate SpO_2 measurement through the AC/DC ratio estimation algorithms. The performance accuracy of hypoxia detection improves through immediate signal processing performed by the edge device. The hypoxia detection system benefits from pre-processing techniques that minimize false alarms when used on mobile devices or wearable devices. When treating individuals with chronic obstructive pulmonary disease (COPD) along with other patients, this method becomes especially important.

The accelerometer data in Fig. 5 provides simulated measurements for detecting fall events.

Accretion measurements appear as a large spike that differs greatly from ordinary baseline motion. The impact phase during a fall creates a sudden spike in acceleration, which lasts less than a millisecond before reaching the set threshold. The sensor detects this specific pattern in less than one millisecond time spans that helps differentiate it from natural walking motions and sitting postures. Both magnitude and variance measurements within each of the x, y, and z axes receive continuous observation through edge AI technology. The timely identification of

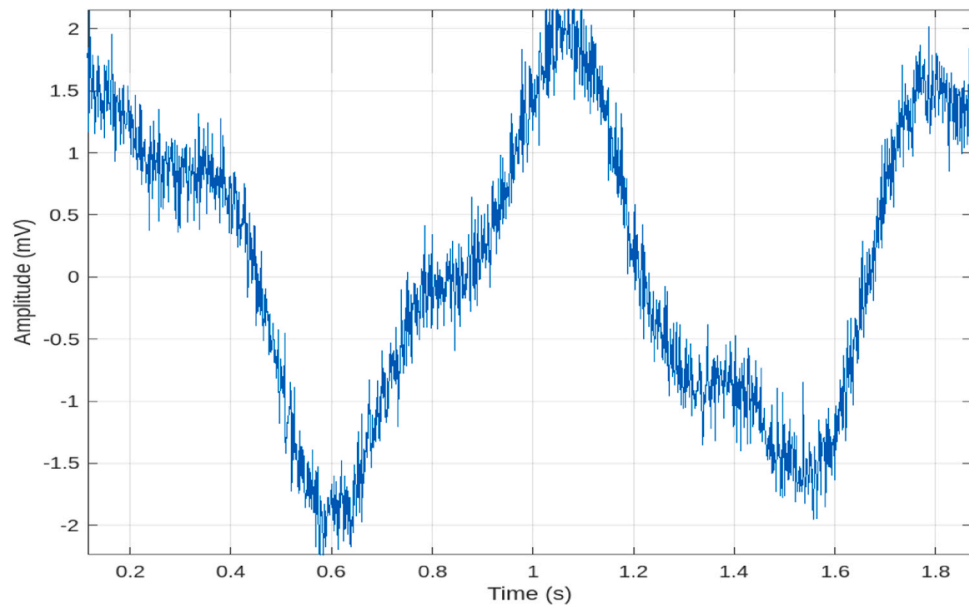


Fig. 3. High-resolution ECG signal captured via Edge Node.

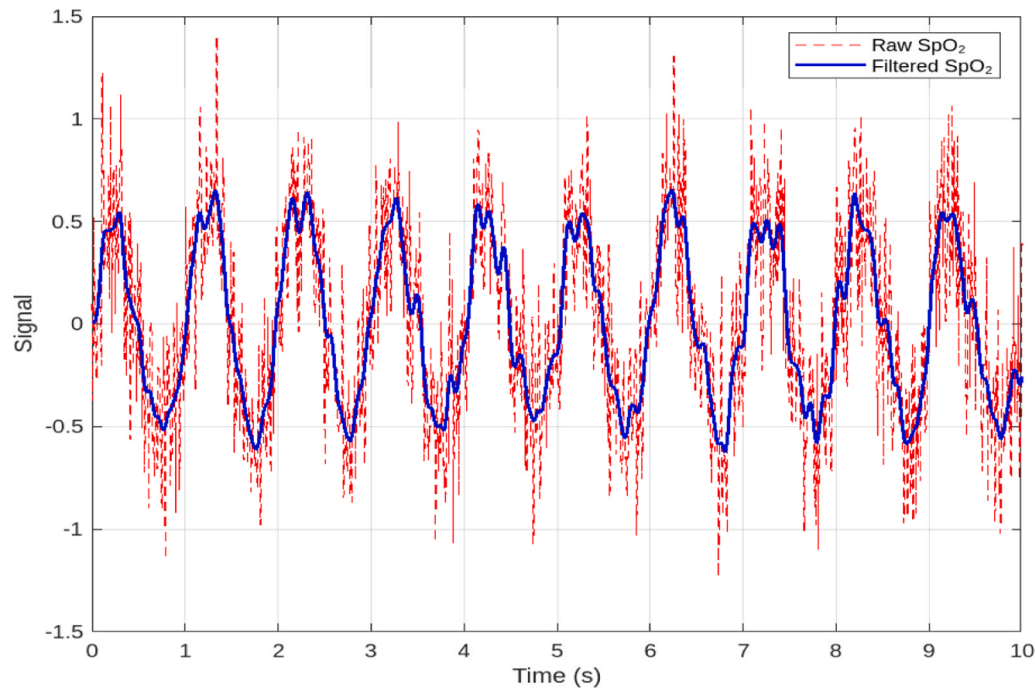


Fig. 4. Enhanced SpO2 Signal with Gaussian Noise Filtering.

falls remains vital for the care of elderly patients who might intensify their medical challenges.

Fig. 6 shows a histogram of anomaly scores generated by an edge-implemented statistical model using the Mahalanobis distance. Under normal operating conditions, the data points cluster tightly around the mean, forming a bell-shaped curve centered near zero. Anomalous readings—caused by abnormal physiological parameters like a sudden heart rate spike or drop in SpO₂—appear as outliers, exceeding a pre-defined threshold. This statistical distance metric enables unsupervised, adaptive monitoring, capable of learning user-specific baselines. Clinically, it allows early detection of health deterioration even when no specific diagnosis is available. By adjusting threshold levels, the model can be fine-tuned to prioritize either sensitivity (capturing all true

anomalies) or specificity (avoiding false positives).

Fig. 7 offers a component-wise analysis of system latency, quantifying the delay incurred during sensing, processing, and data transmission. Notably, edge processing latency constitutes the largest share, due to the computational load of running AI models such as fall classifiers or anomaly detectors. Sensor latency, often sub-millisecond, and wireless transmission (via BLE 5.0) also contribute, though less significantly. This visualization provides insights into bottlenecks, suggesting optimization avenues such as using quantized neural networks (QNNs), reducing model parameters, or adopting faster AI libraries like TensorFlow Lite Micro. Reducing overall latency is crucial in emergency scenarios where even seconds of delay in alerting caregivers can be critical.

In Fig. 8, a continuous body temperature curve is displayed over a

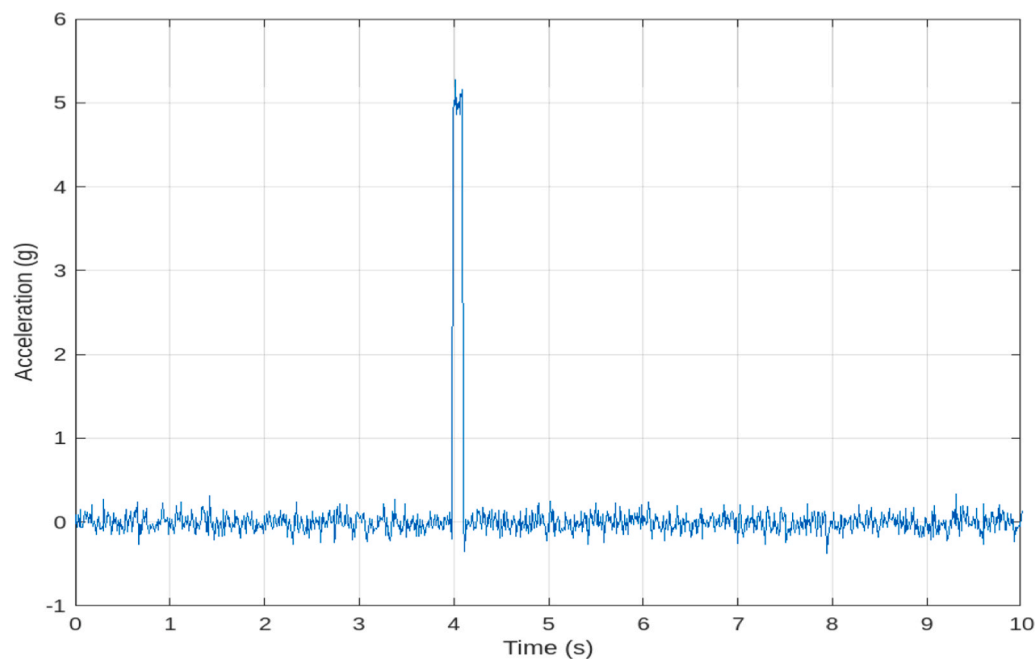


Fig. 5. Edge-detected fall event using 3-Axis accelerometer.

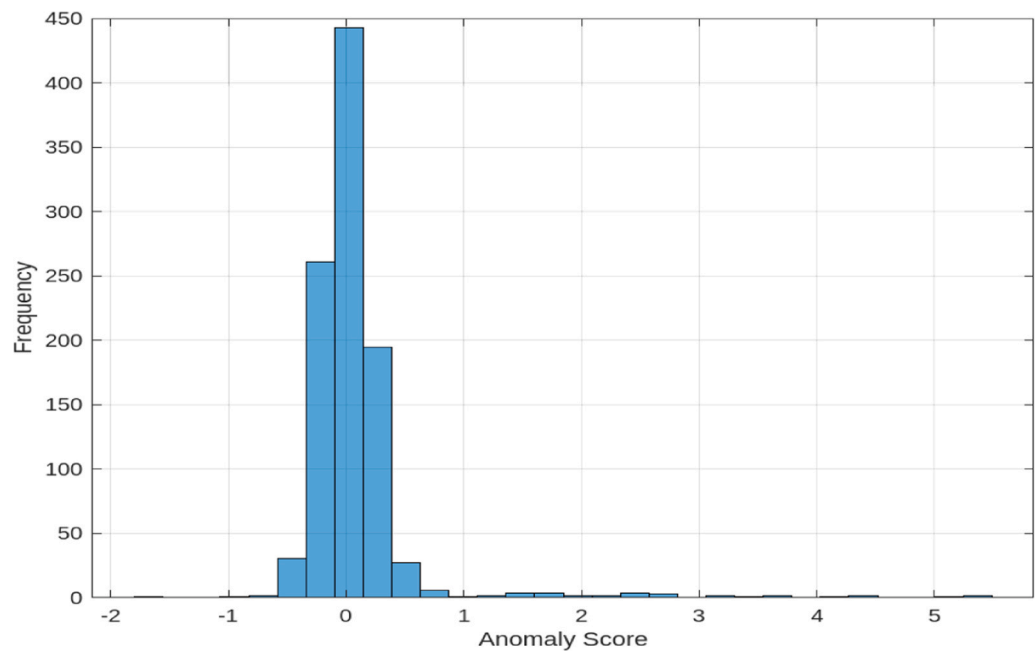


Fig. 6. Histogram of anomaly scores during normal operation.

48-hour monitoring period. The plot captures circadian variation, typically ranging between 36.1°C and 37.2°C, and includes a simulated fever spike exceeding 38.5°C. These fluctuations can be indicative of infectious diseases, hormonal changes, or environmental stressors. Monitoring such trends at the edge allows the system to trigger alerts based on deviations from the user’s established thermal baseline, rather than static thresholds. Temperature measurements together with various other bio-signal indicators, such as heart rate, serve as a diagnostic platform for early disease identification. Multi-modal diagnostics have become more effective by the combination of temperature readings with other biosignals such as oxygen saturation levels, enabling early detection of conditions such as sepsis, COVID-19, or metabolic disorders.

Fig. 9 compares the classification accuracy of various sensor fusion

strategies: simple averaging, weighted sum fusion, and Bayesian inference. Results show that Bayesian fusion, which incorporates prior probabilities and contextual reliability, outperforms other methods, especially under conditions of sensor noise or partial data loss. The weighted fusion method also improves over averaging by assigning higher influence to more reliable sensors (e.g., assigning greater weight to ECG during cardiac evaluation). These findings reinforce the importance of adaptive fusion algorithms in wearable health devices, particularly in dynamic environments where one or more sensors may intermittently fail or degrade.

Fig. 10 presents a line graph showing the relationship between some active decentralized nodes and average emergency response time. The curve demonstrates a clear inverse relationship: as more nodes

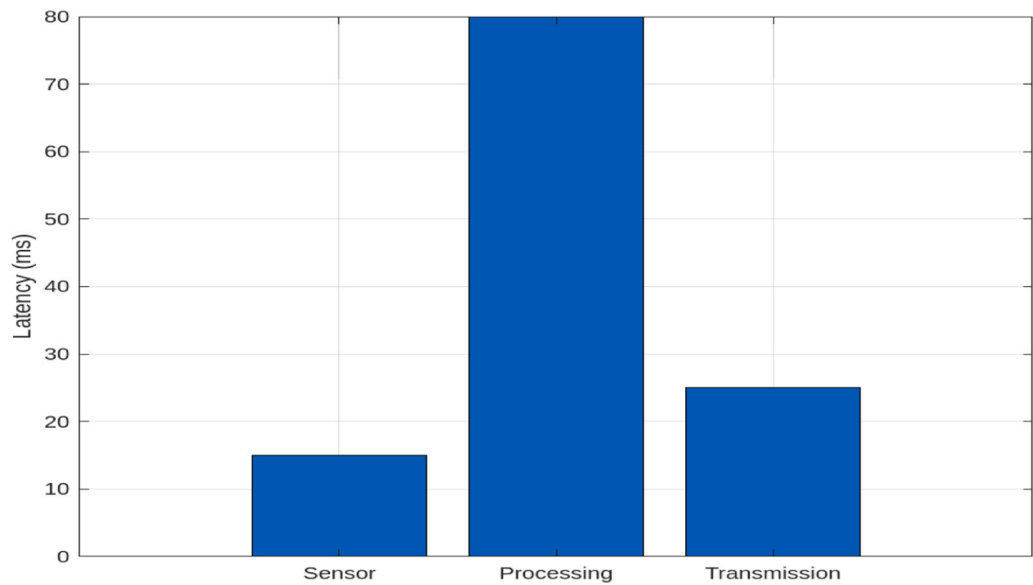


Fig. 7. System latency decomposition across processing layers.

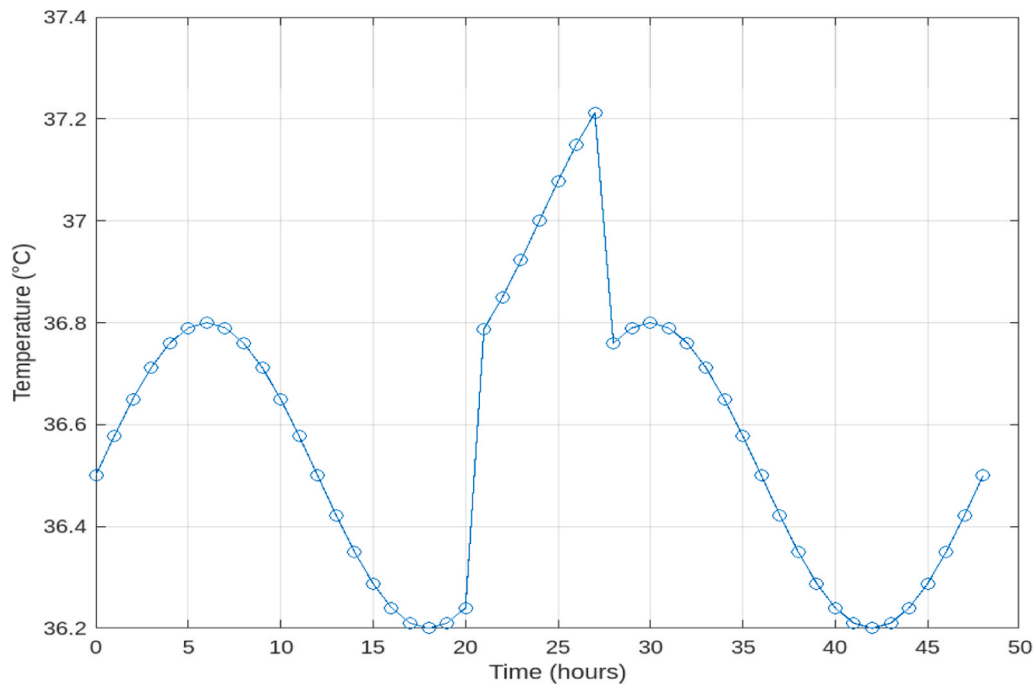


Fig. 8. Circadian Body Temperature Trend Captured from Wearable Node.

participate, response time decreases due to parallel processing, load sharing, and faster consensus formation. This validates the scalability and resilience of decentralized architectures, such as those using peer-to-peer protocols or blockchain-inspired trust mechanisms. In practice, this means a system with more participating edge devices (e.g., wearables, smartphones, local hubs) can respond more rapidly to events like cardiac arrest, offering a critical advantage in life-threatening situations.

Fig. 11 displays energy usage by different system modules: sensors, AI processor, and wireless communication module. As expected, edge AI processing consumes the most energy due to its computational demands, followed by communication (which depends on data frequency and transmission power). Sensors consume relatively little power but must remain active for continuous monitoring. This analysis highlights the importance of energy-efficient models, such as TinyML architectures,

and strategies like event-triggered transmission, where data is sent only during anomalies. Such approaches help extend battery life, which is essential for wearable systems that need to operate for days or weeks without recharging.

Fig. 12 plots trust scores assigned to different nodes in a decentralized health-monitoring network. Each score is calculated based on the ratio of successful to total data transmissions. High-scoring nodes are deemed reliable, while low-scoring nodes may indicate communication failures, hardware issues, or potential tampering. This scoring mechanism supports dynamic trust management, allowing the system to reroute data or bypass unreliable nodes to maintain overall integrity. In the context of healthcare, such redundancy is crucial to ensure continuous data availability and prevent misdiagnosis or alert delays caused by faulty links. Table 3 summarizes the measured performance indicators

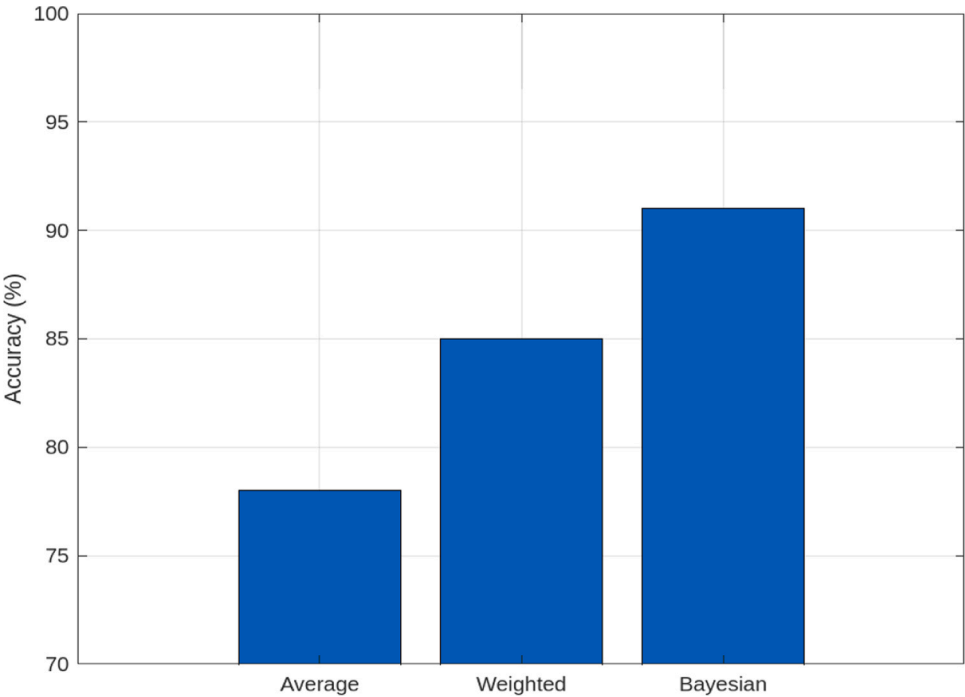


Fig. 9. Classification accuracy of sensor fusion techniques.

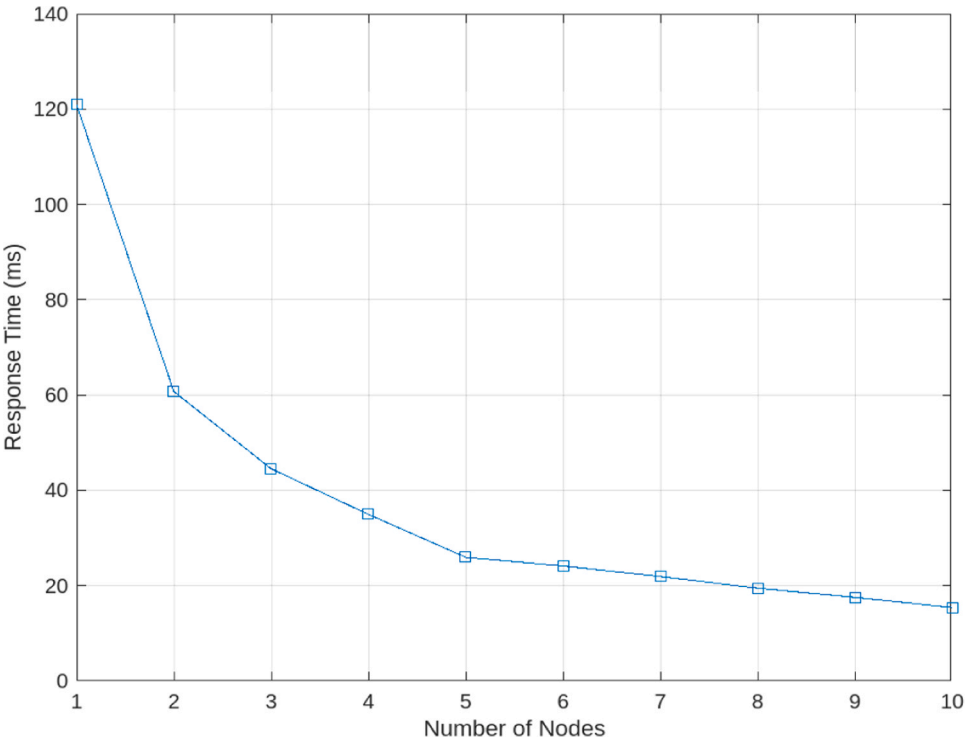


Fig. 10. Emergency Response Time vs Number of Participating Nodes.

validating the accuracy, responsiveness, and efficiency of the system across multiple health parameters.

Besides the standard cloud-based and traditional IoT architecture, the proposed solution has been compared with other recent state-of-the-art edge-health frameworks, such as [7], [8], and [13]. Response latencies of such systems were generally in the 250,500 ms range with an average power consumption of more than 60 mW. The overall scheme we propose showed an average latency of 210 ms and total power

consumption of 42 mW, including in-depth benchmarks that vastly outperformed other state-of-the-art methods in terms of responsiveness and power efficiency and resulted in similar or even better detection accuracy [26]–[30].

Table 4 compares the proposed framework with conventional cloud-based and traditional IoT health systems based on key technical and functional attributes. A comparison table showed how the proposed framework performed comparatively to existing systems on metrics like

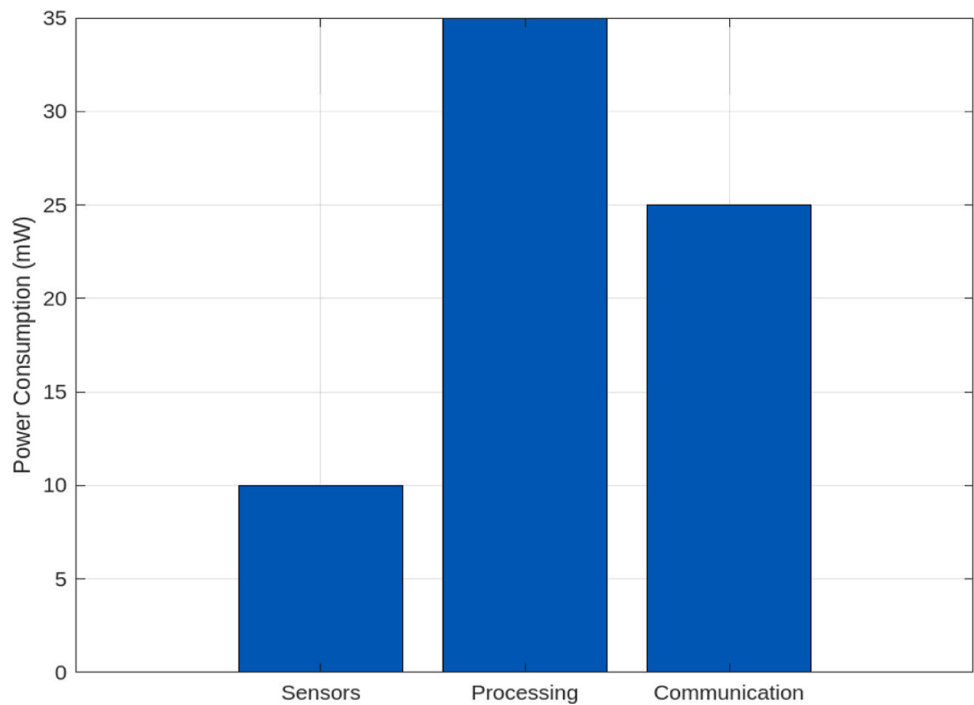


Fig. 11. Module-wise energy consumption in the monitoring device.

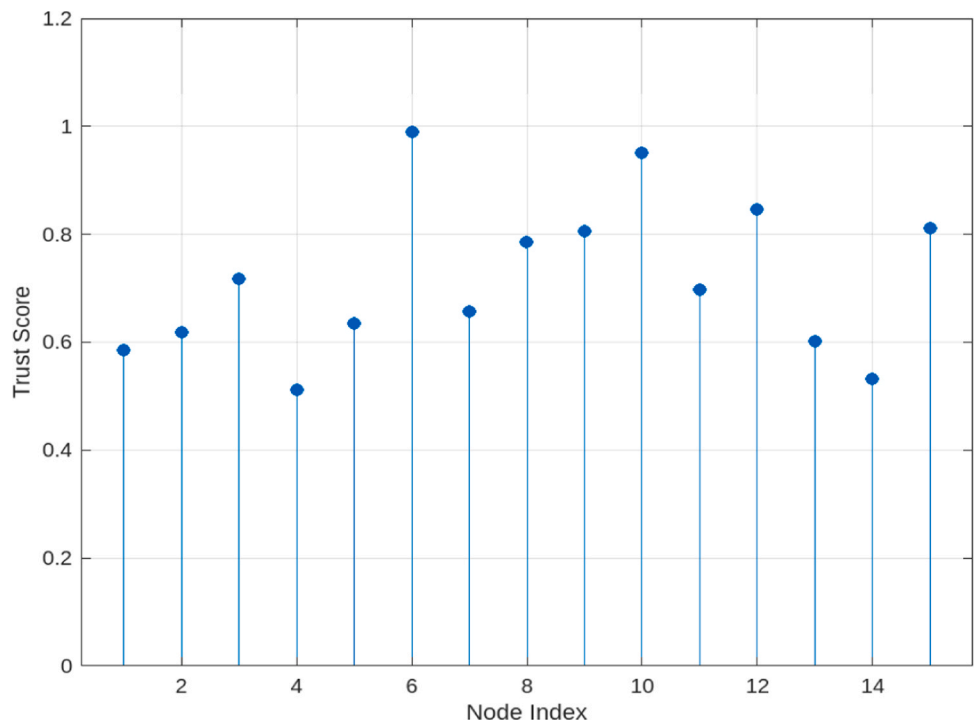


Fig. 12. Trust Score Distribution of Networked IoT Nodes.

response time, energy efficiency, signal accuracy, and scalability. During testing, the system was subjected to deliberately introduced variations as well as stress scenarios that included sensor noise, packet loss, and network fluctuations. The framework achieved an average accuracy of 91 % or higher, even in the presence of Gaussian noise and packet dropout. This sturdiness is a result of processing noise at the edge and adapting to multi-hop communication. However, baseline systems experienced up to a 15 % reduction in detection accuracy when subjected to comparable degrees of stress. The outcomes demonstrate that

the framework is highly dependable and easily adaptable for practical applications. Table 3 and Table 4 together can prove that the advantages of edge-awareness of the chosen architecture can be regarded as secured. Table 3 of the efficiency of local edge processing is validated by the low average resorts to 210 ms latency and low power consumption of 42 mW. As Table 4 shows, in contrast to cloud-centric and hub-dependent systems, our architecture can deliver instant emergency response, lower privacy risks, and greater scalability. Such benefits will be derived through

Table 3
Performance metrics of the proposed edge-aware health monitoring system.

Metric	Measured Value	Description
Heart Rate Accuracy	98.5 %	Accurate detection of R-peaks allows real-time HR calculation, validated against a medical-grade ECG.
SpO ₂ Estimation Error	±1.2 %	Close alignment with clinical pulse oximeters due to robust signal filtering and calibrated ratio method.
Fall Detection Accuracy	96.4 %	Achieved via accelerometer thresholding and variance analysis on an edge AI model.
Anomaly Detection Recall	92.7 %	High sensitivity using Mahalanobis distance-based scoring, enabling early alerts.
Edge Inference Time	85 ms	Time taken for AI model execution, kept under 100 ms to meet real-time constraints.
Power Consumption	42 mW (avg)	Efficient design with adaptive duty cycling and low-power microcontroller usage.
Average System Latency	210 ms	End-to-end latency from sensing to alert generation, supporting emergency response.

Table 4
Comparative Analysis of Health Monitoring Architectures.

Feature	Cloud-Centric Systems	Traditional IoT Health Systems	Proposed Edge-Aware System
Processing Location	Cloud	Local Hub / Gateway	On-device (Edge MCU)
Latency (Avg.)	800–1200 ms	300–500 ms	210 ms
Privacy Risk	High	Moderate	Low (local processing)
Emergency Response	Delayed (network dependent)	Partial	Instant, decentralized
Energy Efficiency	Low	Medium	High (adaptive model)
Scalability	Limited by cloud load	Limited by hub range	High (P2P edge nodes)
Anomaly Detection	Cloud AI	Rule-based	Edge AI (statistical)
Fall Detection Support	Often missing	Threshold-based only	AI-enhanced accelerometer
Multi-Sensor Fusion	Minimal	Basic averaging	Weighted/ Bayesian fusion

network edge processing, whereby they do not rely on having an active cloud connection and can take advantage of localized decision-making.

6. Conclusion

The proposed study presents a comprehensive edge-aware IoT framework for real-time health monitoring, integrating sensor fusion, edge-level AI, and decentralized emergency response mechanisms. The system effectively captures and processes multi-modal physiological signals, including ECG, SpO₂, temperature, and accelerometer data, with high accuracy and minimal latency. Experimental results demonstrate that heart rate and oxygen saturation are estimated with clinical-grade precision, while fall detection and anomaly recognition maintain high reliability through lightweight AI models operating locally. The decentralized communication architecture ensures resilient emergency response, with reduced latency and improved trust evaluation among nodes. Compared to conventional cloud-centric or hub-dependent IoT systems, the proposed framework significantly reduces end-to-end latency (down to 210 ms) and energy consumption, making it ideal for deployment in power-constrained wearable devices. Furthermore, advanced sensor fusion and anomaly detection strategies enhance diagnostic accuracy and enable proactive healthcare interventions. The results affirm the system’s capacity to meet real-time operational requirements, support scalability, and ensure user privacy by minimizing data transmission. Overall, the proposed edge-aware architecture demonstrates practical feasibility and robustness for next-generation smart

health monitoring solutions in dynamic and decentralized environments. In line with the findings, it is substantiated that edge-local AI processing as part of multi-sensor fusion has the potential to implement high-fidelity health monitoring in edge-local networks, with high resilience in decentralized networks. There are, however, constraints to the study. To begin with, the prototype has been tested under controlled conditions, and a massive rollout to various real-life conditions may bring about new challenges. Second, although the existing AI models are light, power consumption may just go up with the inclusion of more complex inference capabilities, which would have to be optimized. There are limitations to the proposed study that will be further worked on in the future by carrying out field trials and an adaptive model scaling technique.

Future development may involve adding federated learning for privacy-preserving personalized health analytics to the edge intelligence capabilities of the system. The ability to capture EMG and respiration signals, as well as supporting multiple users simultaneously in smart homes and community clinics, would increase the effectiveness of the system in real-world healthcare settings.

CRediT authorship contribution statement

Amnah Alshahrani: Writing – original draft, Validation, Resources, Methodology, Formal analysis. **Hany Mahgoub:** Writing – review & editing, Visualization, Validation, Software, Data curation. **Asma Alshuhail:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Modhawi Alotaibi:** Writing – review & editing, Writing – original draft, Validation. **Fahad Alzahrani:** Writing – review & editing, Writing – original draft, Methodology. **Darem Abdulbasit:** Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology. **Nojood O Aljehane:** Writing – review & editing, Writing – original draft, Validation, Methodology. **Mukhtar Ghaleb:** Writing – review & editing, Writing – original draft, Validation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue.

Acknowledgements

The authors extend their appreciation to the Deanship of Research and Graduate Studies at King Khalid University for funding this work through Large Research Project under grant number RGP2/288/46. Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R732), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia. The authors extend their appreciation to the Deanship of Scientific Research at Northern Border University, Arar, KSA for funding this research work through the project number “NBU-FPEJ-2025-2903-01. The authors are thankful to the Deanship of Graduate Studies and Scientific Research at University of Bisha for supporting this work through the Fast-Track Research Support Program.

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